

State Flux and Molecular Deviation

A transport-theoretic extension of the C & G Harmonic with ZNE and ZNY

This chapter defines molecular deviation from a chosen equilibrium state, develops the associated state flux from conservation laws, and then introduces ZNE and ZNY as explicitly dimensionless proposed modifiers. The established transport equations and the proposed harmonic extension are kept conceptually separate.

25.1 Molecular state and reference equilibrium

Let the local molecular state be represented by a state vector containing concentration, flux, density, temperature and any additional variables required by the physical system:

$$\Psi(\mathbf{x}, t) = (C, \mathbf{J}, \rho, T, \dots)$$

Equation (1)

A reference equilibrium concentration field is denoted by $C_{eq}(\mathbf{x}, t)$. The absolute molecular deviation is

$$\Delta C(\mathbf{x}, t) = C(\mathbf{x}, t) - C_{eq}(\mathbf{x}, t)$$

Equation (2)

and the dimensionless relative deviation is

$$\delta_C(\mathbf{x}, t) = \frac{C - C_{eq}}{C_{eq}}$$

Equation (3)

25.2 State flux

For diffusion combined with bulk or geometry-induced transport, the molecular state flux is written as the sum of diffusive and advective contributions:

$$\mathbf{J} = -D\nabla C + C\mathbf{v}$$

Equation (4)

Here D is the molecular diffusivity and $\mathbf{v}$ is the transport velocity. The diffusive term drives matter down the concentration gradient, while the advective term carries matter with the local transport field.

25.3 Conservation law

Local conservation of the dispersed species requires

$$\frac{\partial C}{\partial t} + \nabla \cdot \mathbf{J} = S_C$$

Equation (5)

where S_C represents sources and sinks. Substitution of Equation (4) gives the advection-diffusion equation

$$\frac{\partial C}{\partial t} = \nabla \cdot (D\nabla C) - \nabla \cdot (C\mathbf{v}) + S_C$$

Equation (6)

25.4 Evolution of molecular deviation

Differentiating the deviation field and allowing the reference state itself to vary gives

$$\frac{\partial \Delta C}{\partial t} = \frac{\partial C}{\partial t} - \frac{\partial C_{eq}}{\partial t}$$

Equation (7)

For a stationary equilibrium reference, partial C_{eq} / partial $t = 0$, so

$$\frac{\partial \Delta C}{\partial t} = -\nabla \cdot \mathbf{J} + S_C$$

Equation (8)

For constant diffusivity and incompressible transport, $\text{div}(\mathbf{v}) = 0$, this reduces to

$$\frac{\partial \Delta C}{\partial t} = D \nabla^2 \Delta C - \mathbf{v} \cdot \nabla \Delta C + S_C$$

Equation (9)

25.5 Spherical molecular deviation

For a spherically symmetric field, the radial conservation law is

$$\frac{\partial C}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 J_r) = S_C$$

Equation (10)

$$J_r = -D \frac{\partial C}{\partial r} + C v_r$$

Equation (11)

Combining the two gives

$$\frac{\partial C}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 D \frac{\partial C}{\partial r} \right) - \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 C v_r) + S_C$$

Equation (12)

25.6 Flux deviation

Define the equilibrium flux and the deviation flux by

$$\mathbf{J}_{\text{eq}} = -D \nabla C_{\text{eq}} + C_{\text{eq}} \mathbf{v}$$

Equation (13)

$$\Delta \mathbf{J} = \mathbf{J} - \mathbf{J}_{\text{eq}}$$

Equation (14)

When D and v are unchanged between the actual and reference states,

$$\Delta \mathbf{J} = -D \nabla \Delta C + \Delta C \mathbf{v}$$

Equation (15)

25.7 Incorporation of ZNE and ZNY

Within the proposed harmonic framework, ZNE and ZNY are introduced as dimensionless functions or operators. They must not alter dimensional consistency. Let

$$Z_{NE} = Z_{NE}(\mathbf{x}, t), \quad Z_{NY} = Z_{NY}(\mathbf{x}, t)$$

Equation (16)

The proposed Universal Harmonic state flux is

$$\mathbf{J}_{UH} = Z_{NE}Z_{NY}(-D\nabla C + C\mathbf{v}_G)$$

Equation (17)

where $\mathbf{v}_G$ is the transport velocity associated with the later definition of the G field. Conservation then requires

$$\frac{\partial C}{\partial t} + \nabla \cdot \mathbf{J}_{UH} = S_C$$

Equation (18)

Hence the extended concentration equation becomes

$$\frac{\partial C}{\partial t} = -\nabla \cdot [Z_{NE}Z_{NY}(-D\nabla C + C\mathbf{v}_G)] + S_C$$

Equation (19)

If both modifiers are spatially uniform, they may be taken outside the divergence:

$$\frac{\partial C}{\partial t} = Z_{NE}Z_{NY}[\nabla \cdot (D\nabla C) - \nabla \cdot (C\mathbf{v}_G)] + S_C$$

Equation (20)

25.8 Harmonic equilibrium condition

A zero-net-flux harmonic equilibrium is defined by

$$\mathbf{J}_{UH} = \mathbf{0}$$

Equation (21)

For non-zero ZNE and ZNY, this implies

$$-D\nabla C + C\mathbf{v}_G = \mathbf{0}$$

Equation (22)

$$\nabla \ln C = \frac{\mathbf{v}_G}{D}$$

Equation (23)

If the geometric drift derives from a potential U_G ,

$$\mathbf{v}_G = -\mu_G \nabla U_G$$

Equation (24)

$$C_{eq}(\mathbf{x}) = C_0 \exp\left[-\frac{\mu_G}{D} U_G(\mathbf{x})\right]$$

Equation (25)

Under the Einstein relation $D = \mu_G k_B T$,

$$C_{eq}(\mathbf{x}) = C_0 \exp\left[-\frac{U_G(\mathbf{x})}{k_B T}\right]$$

Equation (26)

25.9 Scalar measure of state-flux molecular deviation

A non-negative scalar diagnostic may be defined from the concentration and flux deviations:

$$D_{\text{mol}}^2 = \left(\frac{\Delta C}{C_{\text{eq}}} \right)^2 + \tau_J^2 \left\| \frac{\Delta \mathbf{J}}{C_{\text{eq}} L} \right\|^2$$

Equation (27)

Here L is a characteristic length and τ_J is a characteristic transport time chosen so that the diagnostic is dimensionless. A ZNE-ZNY weighted form is

$$D_{\text{UH}} = |Z_{\text{NE}} Z_{\text{NY}}| D_{\text{mol}}$$

Equation (28)

25.10 Compact master formulation

$$\Delta C = C - C_{\text{eq}}$$

Equation (29a)

$$\mathbf{J}_{\text{UH}} = Z_{\text{NE}} Z_{\text{NY}} (-D \nabla C + C \mathbf{v}_G)$$

Equation (29b)

$$\frac{\partial C}{\partial t} + \nabla \cdot \mathbf{J}_{\text{UH}} = S_C$$

Equation (29c)

$$\mathbf{J}_{UH} = 0 \iff -D\nabla C + C\mathbf{v}_G = 0$$

Equation (29d)

25.11 Dimensional consistency

Quantity	Meaning	SI units
C	Concentration	mol m ⁻³ or kg m ⁻³
D	Diffusivity	m ² s ⁻¹
J	Molecular flux	mol m ⁻² s ⁻¹ or kg m ⁻² s ⁻¹
v _G	Transport velocity	m s ⁻¹
S _C	Volumetric source	mol m ⁻³ s ⁻¹ or kg m ⁻³ s ⁻¹
Z _{NE} , Z _{NY}	Proposed modifiers	dimensionless

25.12 Scientific status

Equations (1)-(15) are standard continuum transport relations or direct consequences of them. Equations (16)-(29) introduce ZNE, ZNY, the G-associated drift and the molecular-deviation diagnostic as proposed elements of the Universal Harmonic framework. Their physical meaning requires explicit definitions, measurable parameter values and experimental validation.